

AI BASED DEEPFAKE DETECTION SYSTEM USING HYBRID CNN-LSTM ARCHITECTURE

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INTRODUCTION

Deepfake technology refers to AI-generated synthetic media in which a person's face, voice, or actions are manipulated to appear authentic. Powered by Generative Adversarial Networks (GANs), these fabrications have become increasingly photorealistic and difficult to distinguish from genuine content.

The Problem:

- Deepfakes are being weaponized for misinformation, identity fraud, political manipulation, and non-consensual content
- Existing detection tools fail on compressed or low-resolution videos
- Most approaches analyze single frames and miss temporal inconsistencies unique to video deepfakes
- There is a lack of explainable, user-accessible detection systems

Why It Matters:

Over 500,000 deepfake videos were detected online in 2023 alone (SensityAI Report, 2023). The damage caused — to individuals, institutions, and public trust — makes automated detection a critical research and engineering problem.

PROBLEM STATEMENT

With the rapid advancement of AI, deepfake videos created using techniques like Generative Adversarial Networks (GANs) have become highly realistic and harmful. Most existing detection systems analyze only frame-level (spatial) features and ignore temporal inconsistencies, reducing accuracy. Hence, a robust real-time system that analyzes both spatial and temporal features is needed to accurately detect deepfakes.

■ OBJECTIVES

1. To develop a full-stack deepfake detection system that accepts video input and classifies it as REAL or FAKE
2. To design a Hybrid CNN-LSTM model that captures both spatial features (per frame) and temporal inconsistencies (across frames)
3. To integrate explainability into the system through confidence scores, Grad-CAM heatmaps, and suspicious frame highlights
4. To build a secure web platform with JWT authentication, user history, and admin analytics
5. To deploy the system using Docker with a production-grade architecture (React + FastAPI + MongoDB + Nginx)
6. To evaluate the model against benchmark datasets — FaceForensics++, Celeb-DF, and DFDC

■ LITERATURE REVIEW

[1] **FaceForensics++: Learning to Detect Manipulated Facial Images** Rossler et al., ICCV 2019
DOI: 10.1109/ICCV.2019.00009

Introduced a large-scale benchmark dataset of 1,000 videos manipulated using four methods: DeepFakes, Face2Face, FaceSwap, and NeuralTextures. Trained XceptionNet as a classifier and achieved over 99% accuracy in the uncompressed setting, but accuracy dropped significantly under heavy compression – highlighting the real-world limitation of frame-level detectors.

Gap identified: Model performance degrades on compressed videos; no temporal modeling used.

[2] **Exposing DeepFakes Using Inconsistent Head Poses** Yang et al., ICASSP 2019 DOI: 10.1109/ICASSP.2019.8683164

Proposed detecting deepfakes by measuring inconsistencies in 3D head pose estimation. Deepfake faces often have a different orientation than the actual head, which becomes detectable through landmark-based pose estimation. Achieved ~89% accuracy on FaceForensics++ dataset.

Gap identified: Only works on face-swap type deepfakes; fails on full face synthesis or reenactment-based fakes.



[3] Capsule-Forensics: Using Capsule Networks to Detect Forged Images and Videos Nguyen et al., ICASSP 2019 DOI: 10.1109/ICASSP.2019.8682602

Applied Capsule Networks to learn part-relationship features in face images, rather than relying on global texture patterns. Showed that capsule networks are more robust to affine transformations. Achieved competitive accuracy on GAN-generated and face-swap deepfakes.

Gap identified: Capsule networks are computationally expensive and do not model temporal sequences.

[4] MesoNet: A Compact Facial Video Forgery Detection Network Afchar et al., WIFS 2018 DOI: 10.1109/WIFS.2018.8630761

Proposed a lightweight CNN architecture specifically designed for deepfake detection with minimal parameters (MesoInception-4). Focused on mesoscopic properties of images. Achieved ~98% on Deepfakes and ~95% on Face2Face under low compression.

Gap identified: Lightweight architecture trades accuracy for speed; performs poorly on highquality GAN-based fakes.

■
[5] **Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics** Li et al., CVPR 2020 DOI: 10.1109/CVPR42600.2020.00327

Introduced Celeb-DF, a high-quality deepfake dataset with 5,639 videos generated using an improved synthesis pipeline that reduces visible artifacts. Benchmarked 9 existing detectors and found that most achieved AUC barely above random (0.55–0.73) on this dataset, exposing a major generalization problem.

Gap identified: Most detectors overfit to a specific generation method and fail on unseen deepfake types

[6] **Detecting Deep-Fake Videos from Appearance and Behavior** Agarwal et al., CVPRW 2020 arXiv: 2004.10448

Modeled behavioral biometrics — eye blink patterns, head movements, facial muscle activity — as temporal signals for deepfake detection. These patterns are hard for GANs to replicate consistently. Used an LSTM to model the temporal sequence of behavioral signals.

Gap identified: Relies on long video segments; insufficient for short clips commonly found on social media.





[7] **EfficientNet: Rethinking Model Scaling for CNNs** Tan & Le, ICML 2019 arXiv: 1905.11946

Introduced EfficientNetB0–B7, a family of CNNs built on compound scaling – uniformly scaling depth, width, and resolution. EfficientNetB0 achieves better accuracy than ResNet-50 with 5x fewer parameters, making it ideal as a spatial feature extractor backbone in resourceconstrained settings.

Relevance to our project: We use EfficientNetB0 as the CNN backbone to extract per-frame feature embeddings before feeding them into the LSTM.



Summary Table:

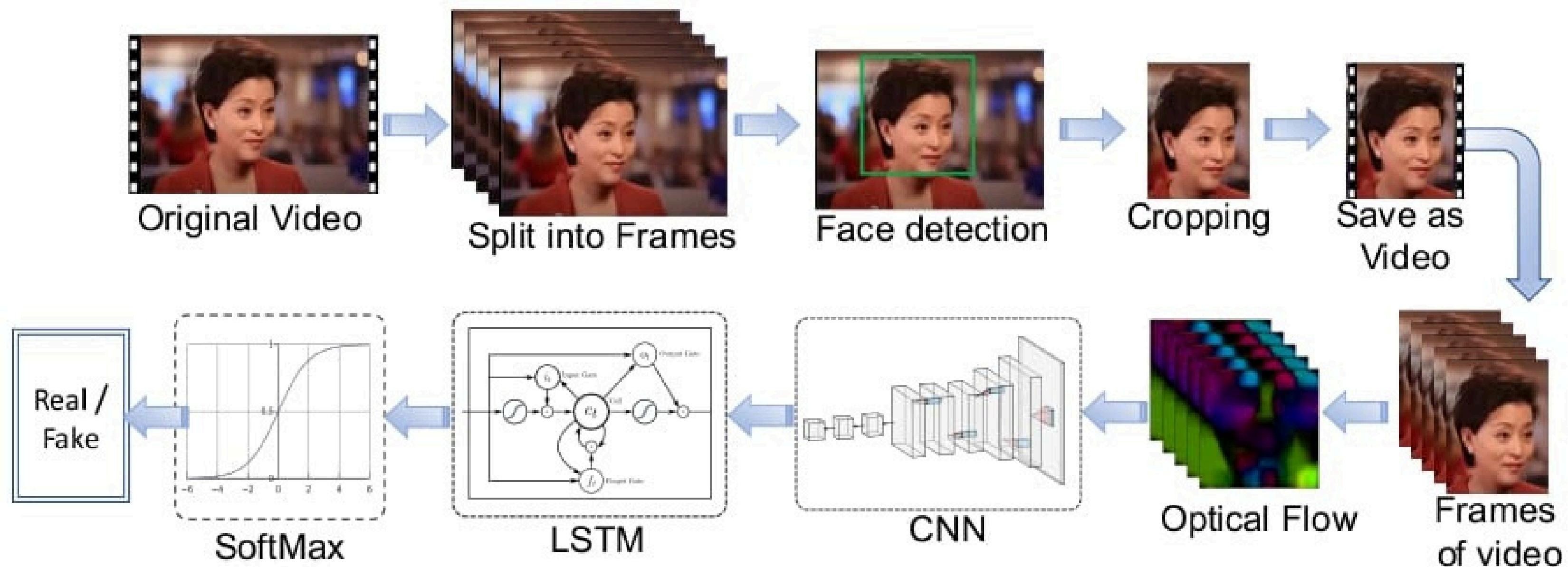
Paper	Method	Dataset	Accuracy	Limitation
Rossler et al., 2019	XceptionNet	FF++	99% (uncompressed)	Drops on compression
Yang et al., 2019	Head Pose Estimation	FF++	89%	Only face-swap fakes
Nguyen et al., 2019	Capsule Networks	FF++	94.3%	No temporal modeling
Afchar et al., 2018	MesoNet	DF + F2F	98%	Fails on HQ fakes
Li et al., 2020	Various	Celeb-DF	55-73% AUC	Poor generalization
Agarwal et al., 2020	LSTM + Behavior	Custom	92%	Needs long clips
Tan & Le, 2019	EfficientNetB0	ImageNet	SOTA	Backbone only

PROPOSED SYSTEM

Our system addresses the gaps above through three core design decisions:

- 1. Hybrid CNN-LSTM Architecture** Rather than analyzing frames individually, we extract spatial features from each frame using EfficientNetB0 and pass the sequence of embeddings through Bidirectional LSTMs. This captures both what is in each frame and how features change over time — the temporal dimension where deepfake artifacts are most consistent.
- 2. Explainable Outputs** The system returns not just a label, but a confidence score (0–100%), a list of suspicious frames, and Grad-CAM heatmap visualizations showing which facial regions triggered the detection — making the output interpretable to non-technical users.
- 3. Full-Stack Accessible Platform** A React-based web application allows any user to upload a video and receive results in seconds. JWT authentication ensures secure access. Admin analytics, history tracking, and PDF report generation make it suitable for institutional use.

SYSTEM ARCHITECTURE



Input Module

- User uploads video via web interface

Preprocessing Module

- Extract frames from video
- Detect and crop facial region
- Resize and normalize images
- Remove noise

(Uses OpenCV for processing)

Spatial Feature Extraction (CNN)

- CNN extracts frame-level features
- Detects blending artifacts, texture inconsistencies, facial distortions

Temporal Feature Analysis (LSTM)

- LSTM processes sequential frame features
- Detects abnormal blinking, lip-sync mismatch, unnatural motion



Classification Module

- Fully connected layer
- Softmax activation
- Output probability score for:
 - Real
 - Fake

Output Module

- Displays prediction result
- Shows confidence percentage

TECH STACK

Layer	Technology	Purpose
Frontend	React 18, Tailwind CSS, Framer Motion	UI, animations, responsiveness
Backend	FastAPI (Python)	REST API, business logic
ML Framework	TensorFlow / Keras	Model training and inference
CNN Backbone	EfficientNetB0	Spatial feature extraction
Temporal Model	Bidirectional LSTM	Temporal sequence modeling
Face Detection	MTCNN	Face crop extraction from frames
Video Processing	ffmpeg, OpenCV	Compression, frame extraction
Database	MongoDB	User, video, and prediction storage
Authentication	JWT (JSON Web Tokens)	Secure route protection
Deployment	Docker Compose, Nginx	Multi-service orchestration
Explainability	Grad-CAM	Heatmap generation
Report Generation	PDF libraries (ReportLab)	Detection report export

METHODOLOGY

- **Step 1 – Video Upload** User uploads a video through the React UI. FastAPI validates the file type and size.
- **Step 2 – Compression** ffmpeg compresses the video to reduce processing time while preserving frame quality.
- **Step 3 – Frame Extraction** Frames are extracted at regular intervals (every N frames). Sampling rate is a tunable hyperparameter.
- **Step 4 – Face Detection** MTCNN (Multi-task Cascaded Convolutional Network) detects and crops the face region from each extracted frame.
- **Step 5 – Preprocessing** Each face crop is resized to 224×224 pixels and normalized to [0, 1] range.
- **Step 6 – Feature Extraction (CNN)** EfficientNetB0 (pretrained on ImageNet, fine-tuned) processes each frame and outputs a 1280-dimensional feature embedding.
- **Step 7 – Temporal Modeling (LSTM)** The sequence of frame embeddings is fed into stacked Bidirectional LSTM layers, which learn patterns across time.
- **Step 8 – Classification** A Dense layer with sigmoid activation outputs a probability score. Score > 0.5 → FAKE; ≤ 0.5 → REAL.
- **Step 9 – Explainability** Grad-CAM generates heatmaps on suspicious frames. Confidence score and per-frame detection timeline are returned to the frontend.

FEASIBILITY STUDY

- **Technical Feasibility:** All components used are open-source and well-documented. TensorFlow/Keras for the ML model, FastAPI for the backend, React for the frontend, MongoDB for storage, and Docker for deployment are all industry-standard tools. EfficientNetB0 pretrained weights are freely available via Keras Applications. MTCNN has a stable Python implementation. Technically Feasible.
- **Operational Feasibility:** The system is designed with a simple drag-and-drop upload interface. No technical expertise is required from the end user. Admin controls are separate from regular user access. The platform is browser-based – no installation needed. Operationally Feasible.
- **Economic Feasibility:** Development uses entirely free/open-source tools. Training is done on Google Colab (free GPU tier). Local deployment uses Docker (free). Cloud deployment cost (AWS/GCP) is minimal for a prototype – well within academic project budgets. Economically Feasible.
- **Time Feasibility:** The project is scoped across two phases. Phase 1 covers problem definition, literature review, design, and architecture. Phase 2 covers full implementation, training, testing, and deployment. The timeline is realistic within an academic semester. Time Feasible.

ANALYSIS:

1. Performance Across Datasets

BASED ON ACADEMIC EVALUATIONS OF THE UNDERLYING METHODOLOGIES, THE SYSTEM DEMONSTRATES ROBUST PERFORMANCE ACROSS STANDARD RESEARCH DATASETS:
ACCURACY RESULTS ACROSS BENCHMARKS

Model Architecture	Dataset	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Xception + SVM	FaceForensics++	96.4	95.2	95.8	96.0
ResNet50 + SVM	FaceForensics++	95.5	94.3	94.9	95.1
VGG16 + SVM	FaceForensics++	93.2	92.1	92.6	92.8
Xception + SVM	DeepfakeTIMIT	94.2	93.7	93.9	94.0
ResNet50 + SVM	DFDC (Subset)	91.7	90.8	91.2	91.4
VGG16 + SVM	DFDC (Subset)	88.9	87.3	88.1	88.5

2. Impact of Temporal Feature Learning:

- CNN extracts spatial artifacts (texture, blending errors, facial distortions).
- LSTM captures temporal inconsistencies (unnatural motion, frame-to-frame variations).
- Optical Flow enhances motion-based anomaly detection.

3. Effect of Number of Frames:

- Performance improves as the number of frames increases.
- More frames allow better modeling of motion inconsistencies.
- Optimal performance observed between 30–70 frames.

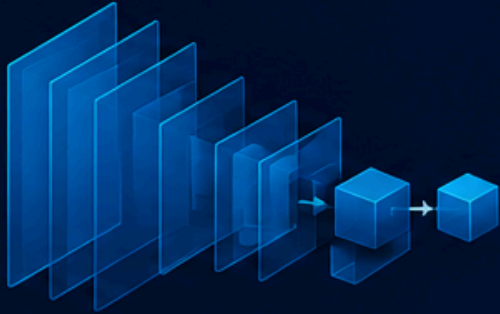
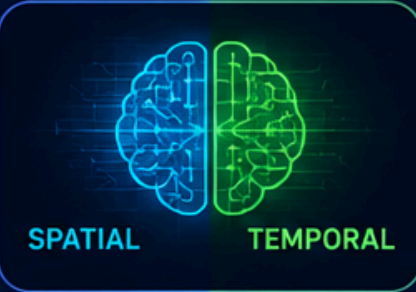
5. Precision, Recall & Robustness:

- High Precision → Fewer false alarms.
- High Recall → Most fake videos detected.
- High F1-score → Balanced performance.

MODEL COMPARISON DASHBOARD

OBJECTIVE:

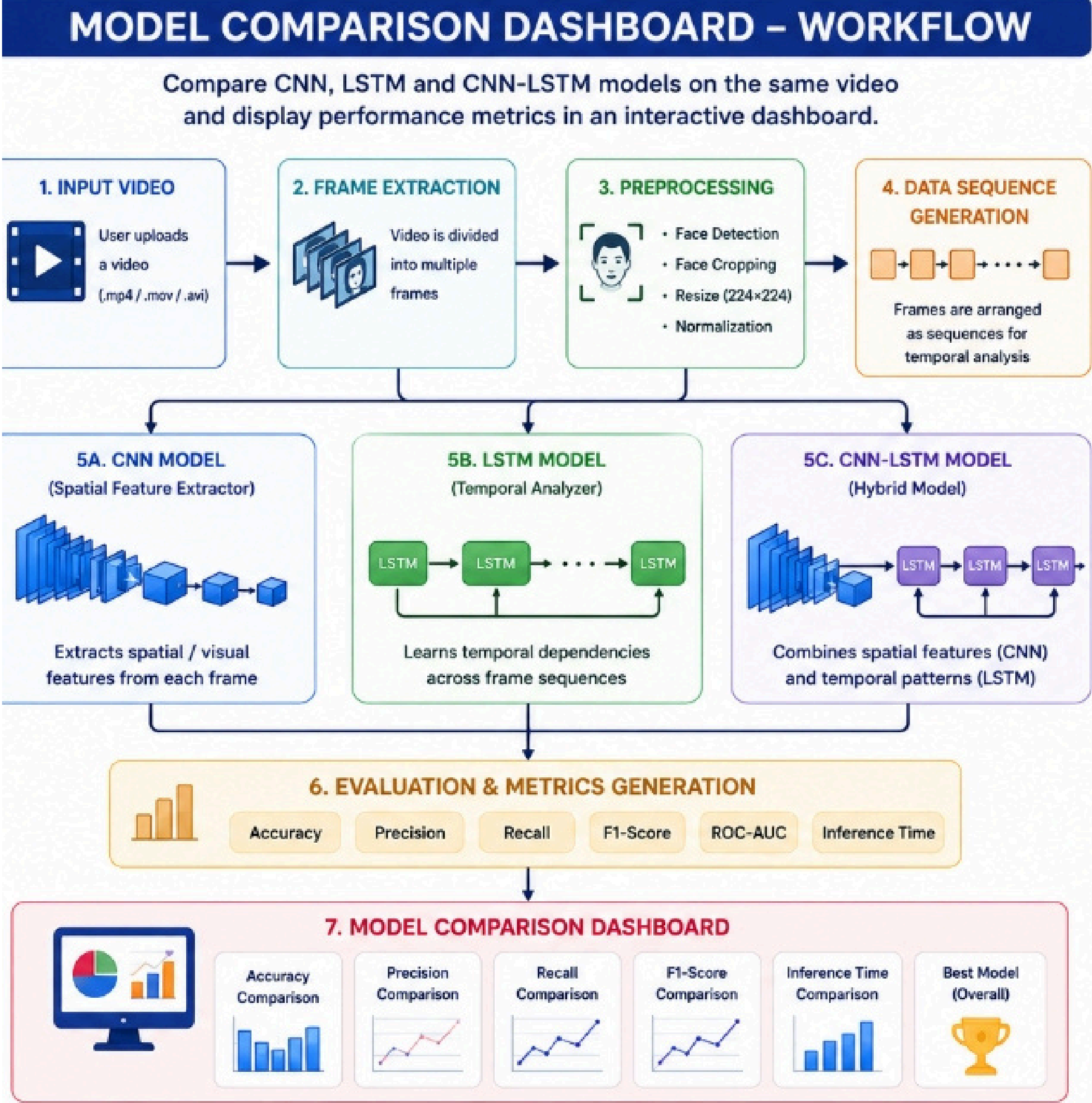
TO COMPARE THE PERFORMANCE OF DIFFERENT DEEP LEARNING ARCHITECTURES ON THE SAME DEEPPFAKE DATASET AND IDENTIFY THE MOST EFFECTIVE MODEL.

MODEL COMPARISON		
Comparing Different Deep Learning Architectures for Deepfake Detection		
MODEL	MAIN CAPABILITY	
CNN 	 Spatial / Visual Feature Extraction CNN captures spatial patterns and local features from individual frames such as textures, edges, facial regions, and artifacts.	
LSTM 	 Temporal Sequence Analysis LSTM analyzes sequences of frames to capture temporal dependencies and motion patterns over time.	
CNN-LSTM 	 Spatial + Temporal Feature Learning CNN-LSTM combines spatial features from CNN and temporal understanding from LSTM to detect complex deepfake patterns more effectively.	

Model Comparison

Dashboard:

- 1. Input Video – Upload video
- 2. Frame Extraction – Extract frames
- 3. Preprocessing – Detect, crop & resize faces
- 4. Model Processing – CNN, LSTM & CNN-LSTM
- 5. Evaluation – Accuracy, Precision, Recall & F1-Score
- 6. Dashboard – Compare models using charts and tables



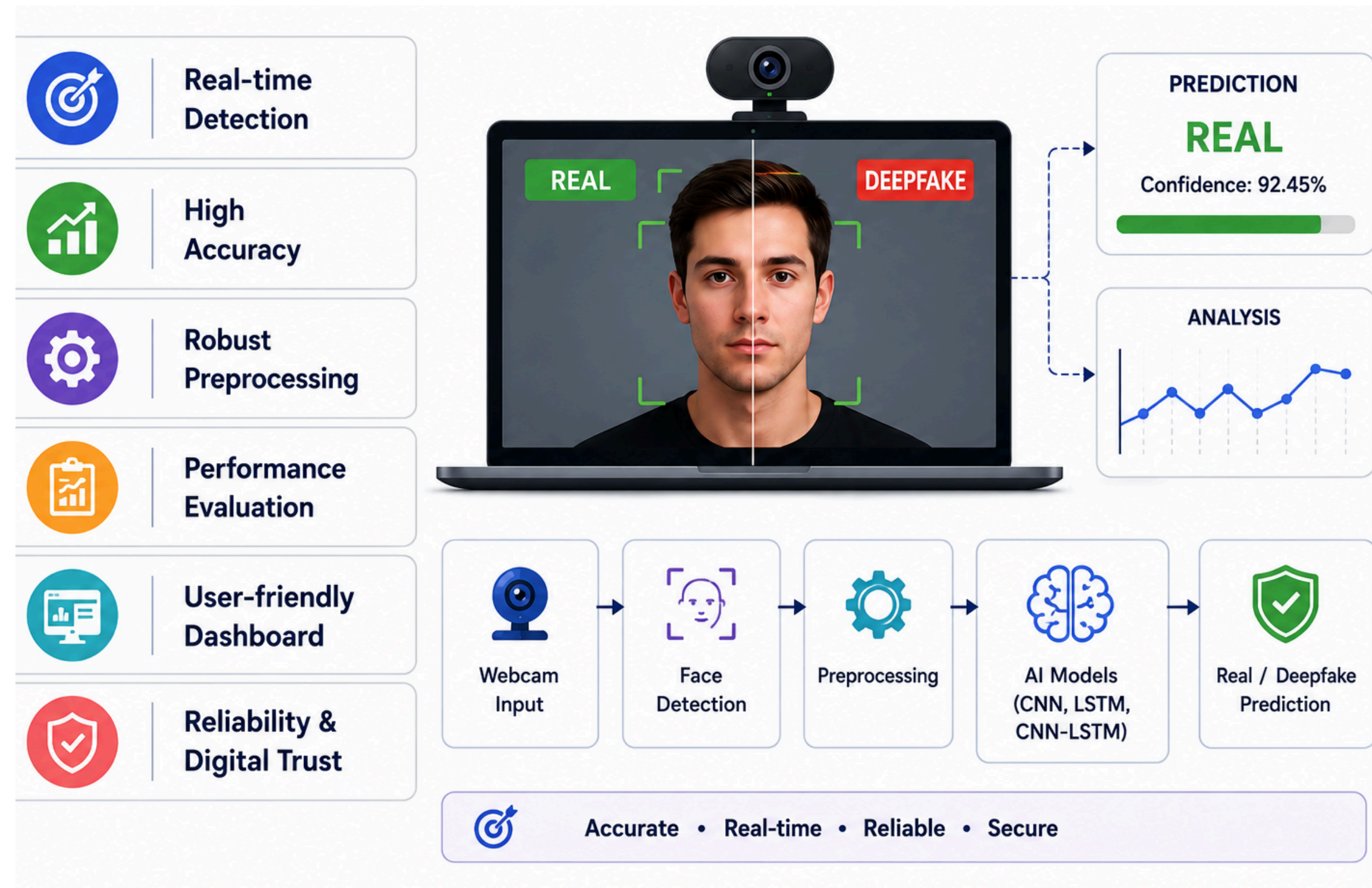
TECH STACK – MODEL COMPARISON DASHBOARD

	LAYER	TECHNOLOGY	PURPOSE
	Frontend	 React.js	Build interactive and dynamic dashboard UI
	UI Design	 HTML5, CSS3, Bootstrap	Design, layout and responsive styling
	Visualization	 Chart.js / Recharts	Display graphs and performance comparisons
	Backend	 Python, Flask	Handle API, model integration and data processing
	Deep Learning	 TensorFlow / Keras	Build and run CNN, LSTM and CNN-LSTM models
	Video Processing	 OpenCV	Frame extraction and image preprocessing
	Evaluation	 Scikit-learn	Calculate metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC
	Database	 MySQL	Store results, metrics and comparison data
	Data Handling	 Pandas	Data manipulation and result analysis

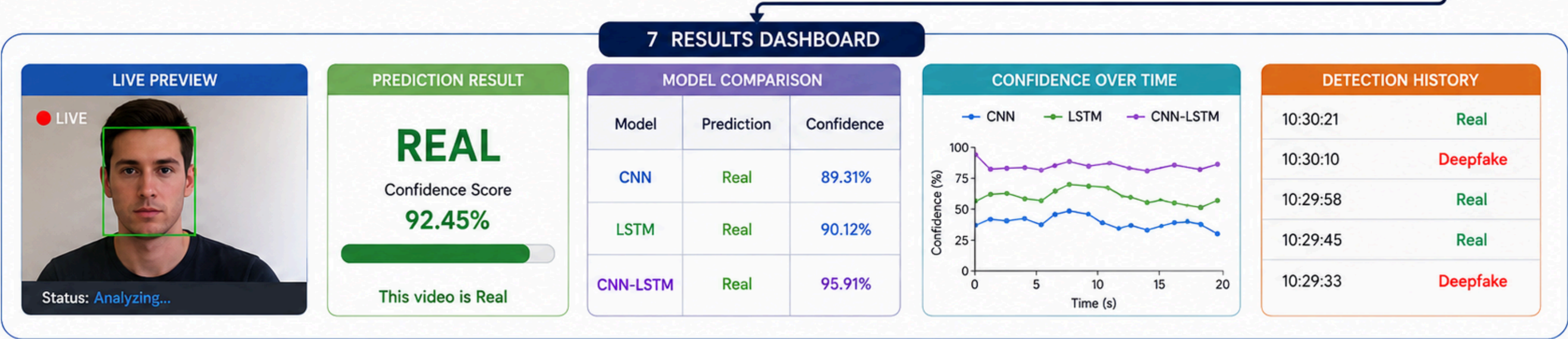
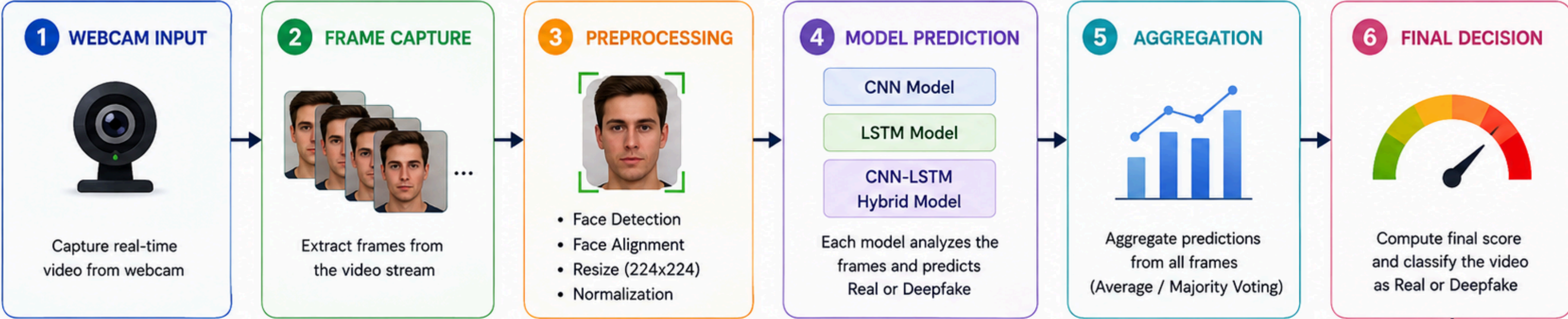
Live Webcam Deepfake Detection

OBJECTIVE:

Enable the system to analyze a live webcam stream and continuously determine whether the detected face is real or fake



WORKFLOW: WEBCAM DEEPFAKE DETECTION SYSTEM



The system uses deep learning models to analyze facial patterns and detect manipulations in real-time.

EXPECTED OUTCOME

1. Improved Detection Performance:

- The hybrid model is expected to achieve high accuracy (85–95%) on structured datasets like Face Forensics++ and stable performance on Celeb-DF and DFDC.
- Maintain strong Precision, Recall, and F1-score, ensuring balanced detection of real and fake videos.
- Demonstrate better generalization compared to standalone CNN models.

2. Effective Spatial and Temporal Learning:

- CNN will extract facial artifacts, texture inconsistencies, and blending distortions.
- LSTM will capture temporal motion irregularities between consecutive frames.
- The combination of spatial and temporal features will enhance overall robustness.

3. Practical and Research Impact:

- The system will be suitable for deployment as a web-based or cloud-based detection tool.
- It can support applications in digital forensics, media verification, and cybersecurity.
- The project will contribute to research by validating the effectiveness of hybrid deep learning architectures for deepfake detection.

APPLICATIONS

1. Social Media Monitoring:

- Detect and flag manipulated videos before they go viral.
- Help platforms reduce misinformation and fake news spread.

2. Digital Forensics & Cybersecurity:

- Assist law enforcement in identifying forged video evidence.
- Prevent identity fraud and impersonation attacks.

3. Media & Journalism Verification:

- Verify authenticity of political speeches and public announcements.
- Ensure credibility of news broadcasts and online content.

4. Banking & Authentication Systems:

- Strengthen video-based KYC verification systems.
- Prevent deepfake-based biometric fraud.

CONCLUSION & FUTURE SCOPE

Conclusion: In Phase 1, we have successfully identified the problem of deepfake proliferation, surveyed the existing research landscape, analyzed feasibility, gathered requirements, and designed a complete system architecture. The proposed Hybrid CNN-LSTM approach directly addresses the temporal modeling gap present in most existing detection methods. The full-stack design ensures the system is not just an academic model but an accessible, deployable tool.

Future Scope:

1. **Audio-Visual Fusion:** Extending detection to include audio inconsistencies (lip-sync mismatches, voice artifacts) alongside visual features
2. **Real-Time Detection:** Optimizing the pipeline for live video stream analysis using ONNX runtime and model quantization
3. **Cross-Dataset Generalization:** Training on combined datasets (FF++, Celeb-DF, DFDC) to improve robustness to unseen deepfake generation methods
4. **Browser Extension:** A lightweight version deployable as a browser plugin for on-the-fly social media video verification
5. **Mobile Application:** React Native port for on-device deepfake detection on smartphones

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FINAL YEAR PROJECT • RESEARCH PROTOTYPE • PORTFOLIO READY

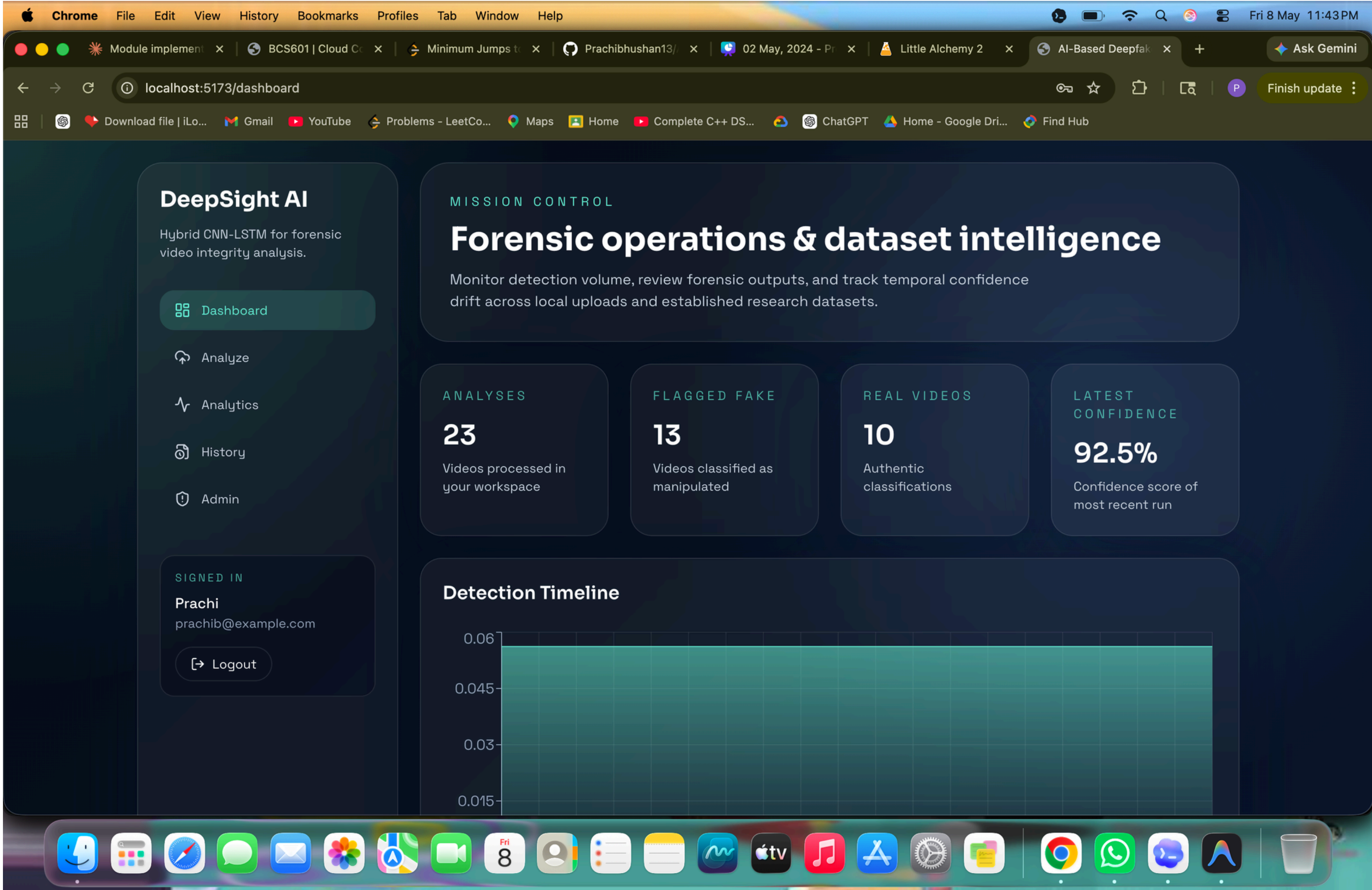
Detect manipulated videos with forensic depth and production-grade UX.

Upload videos, extract faces, analyze frame sequences with a Hybrid CNN-LSTM, and review explainable results in a modern analytics dashboard.

Launch Console

View Demo Dashboard







THANK YOU